MID-SEMESTER EXAMINATION, April-2025
Introduction To Databases (CSE-3151)

Program: B.Tech (CSE & CSIT) | **Semester: 6th**
Full Marks: 30 | **Time: 2 Hours**

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
To identify and explain the different components and functionalities of DBMS and their interdependence through the database architecture.	L1, L2	1a, 1b, 1c, 2a	8
To analyze an enterprise schema for given user requirements and apply the conceptual database design principles through ER modelling to construct the ER diagram and corresponding relation schemas.	L4	2b, 2c 3a, 3b, 3c	10
To analyze and design relational database schema using decomposition and normalization techniques.	L5, L6	4a, 4b, 4c 5a, 5b, 5c	12

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) List four significant differences between a file-processing system and DBMS. 2

   (b) Explain the differences between 2-tier and 3-tier architectures. Which is better suited for web applications? 2

   (c) What is the need of data abstraction in a database system? Further explain different levels of abstraction. 2

2. (a) What are the different types of database users? How do their roles impact the overall database system? 2

   (b) Explain simple, composite, multivalued and derived attributes with suitable examples. 2

(c) What is the difference between overlapping and disjoint specialization? Develop the appropriate relational schemas for the given diagram in figure 1. 2

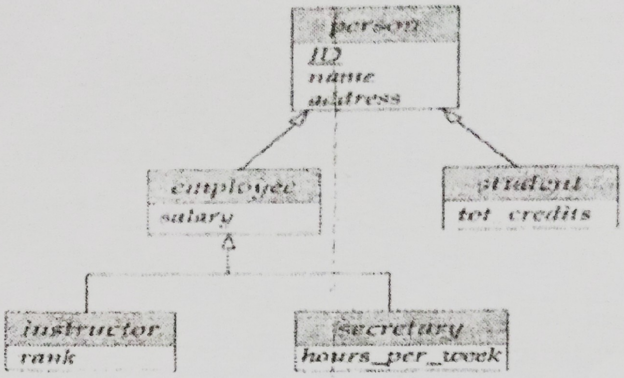


Figure 1.

3. (a) Consider a database maintaining the details of the orders placed by customers in an online shopping system. Each customer is represented by a unique id, name, phoneno, address. Customers may have more than one phoneno, and address includes the city, state and pincode. One customer can place many orders. Each order is represented by a unique order_no, amount, date and order_status. Order includes some products. Each product is represented by a unique product_no, product_name, description and price. The association of customer and order is represented by relationship named as "places" and association of order and product is represented by the relationship name "includes". One order can include many products and many orders can be placed for the same product. The database also keeps information about the customers who have registered but have not placed any order. Construct the ER diagram for the above database representing entity set, relationship set, mapping cardinality and participation constraints. 2

(b) Develop the appropriate relational schemas for the database of online shopping system illustrated in the ER diagram related to que. 3(a). 2

(c) Analyze the referential integrity for the relational schema resulted in the que. 3(b) and construct its corresponding schema diagram. 2

4. (a) Consider the following set F of functional dependencies on the relational schema R(A, B, C, D, E, F), F={A→BCD, BC→DE, B→D, D→A}. Compute the closure of B. Prove using Armstrong's axiom that AF is a super key. 2

(b) Consider the following table T, and the set of functional dependencies F: {A→B, C→B, D→ABC,AC→D} 2

A	B	C	D
a1	b1	c1	d1
a1	b1	c2	d2
a2	b1	c1	d3
a2	b1	c3	d4

Given the following records or tuples, indicate which record can be added in to table T without violating any of the functional dependencies in F. If a record or tuple cannot be legally added, justify it indicating which functional dependency is violated.
Records: (a2,b1,c4,d8), (a1,b1,c2,d5), (a3, b1, c4, d3), (a2, b2, c1, d8).

(c) Consider two sets of FDs, F and G, F = {A → B, B → C, AC → D} and G = {A →B, B → C, A → D} Are F and G equivalent? Find Canonical cover of F. 2

5. (a) Consider a schema R(A,B,C,D) with the functional dependencies on the relation schema: 2
F = {A→B and C→D}. Check both the properties of decomposition if R is decomposed into R1(A,B) and R2(C,D). If any of properties are not satisfied then what should be the actual decomposition of the relation schema R.

(b) Consider a relational schema 2
FinalyearProject(student, researchTopic, consultationDay, supervisor) with the functional dependencies:
F = { student → researchTopic,
student, researchTopic→supervisor,
researchTopic, supervisor → consultationDay}.
Which is the weakest normal form that the schema satisfies? Comment whether it satisfies the property of 3NF or not with reason. If no, redesign the schema satisfying the properties of 3NF. **(GATE 2019)**

(c) Consider the relational schema StudentPerformance(Name, CourseNo, RegNo, Grade) has the following functional dependencies: 2
F = {Name,CourseNo → Grade,
RegNo ,CourseNo → Grade,
Name→ RegNo,
RegNo→ Name}
Which is the highest normal form of this relation schema satisfies? **(GATE 2004)**

End of Questions

END-SEMESTER EXAMINATION, May-2025

Introduction to databases (CSE-3151)

Programme: B.Tech (ALL CSE & CSIT) **Semester: 6th**
Full Marks: 60 **Time: 3 Hours**

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
To identify and explain the different components and functionalities of DBMS and their interdependence through the database architecture.	L1, L2	1a,b,c	6
To formulate SQL queries and able to develop a database application as per user requirements using SQL and database connectivity.	L3	7b	2
To analyze an enterprise schema for given user requirements and apply the conceptual database design principles through ER modelling to construct the ER diagram and corresponding relation schemas.	L2, L4	2a,c 3a,b,c	10
To analyze and design relational database schema using decomposition and normalization techniques.	L2,L5, L6	2b 4a,b,c 5a,b,c	14
To apply relational algebra and relational calculus to express queries on relational schemas.	L4	6a,b,c 7a,c	10
To interpret the functional issues related to transaction and database recovery along with the concept of storage and database system architecture.	L4	8a,b,c 9a,b,c 10a,b, c	18

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) What is a data model? Differentiate between relational and semi-structure data model. — 2

   (b) Explain the roles of the query processor and the storage manager in database architecture. — 2

   (c) How DDL and DML statements affect instances and schemas of a database? — 2

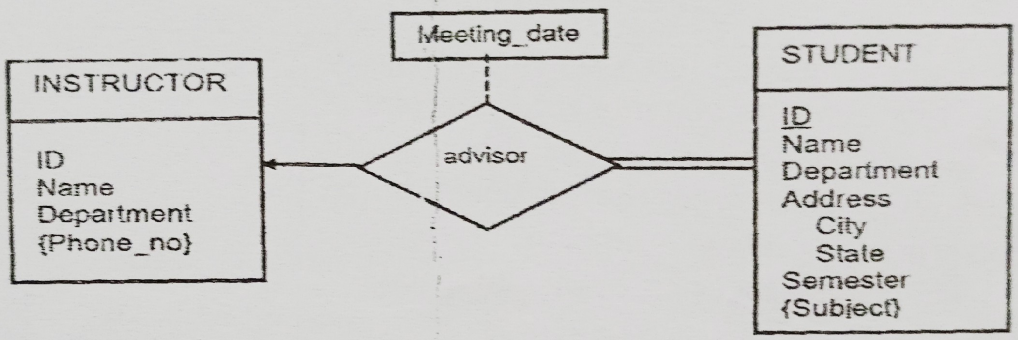
2. (a) Discuss the binary and ternary degree of relationships in the ER model with suitable examples. 2

(b) Consider a relational schema R = {E,F,G,H,I,J,K,L,M,N} and the set of functional dependencies EF→G, F→IJ, EH→ KL, K→M, L→N on R. Find the candidate key and at least one super key of R. **(GATE 2014)** 2

(c) Distinguish between attribute inheritance and participation inheritance in specialization with example. 2

3. (a) An entity set order with attributes <u>order_num</u>, order_date and payment_info, and another entity set order_item with attribute item_num, quantity and price are associated with a relationship set "contains". The existence of order_item is dependent on the order. Construct the suitable ER Diagram to represent this scenario. 2

(b) Convert the given part of ER diagram into minimum number of relational schemas. 2



(c) Analyze the relational schema given in 3(b), and draw its corresponding schema diagram. 2

4. (a) Explain First Normal Form (1NF) with a suitable example. Why 1NF is considered a foundational step in normalization? 2

(b) Find the canonical cover of the following set F of functional dependencies on the schema R(A,B,C), 2
F = {A→BC, B→C, A→B, AB→C}.

(c) What is partial dependency? Identify the number of partial dependencies in the relational schema R(A,B,C,D,E) with FD set 2
F = {A→BE, E→C, C→AB}.

5. (a) Consider the relation schema R(A, B, C, D,E) with functional dependency set F = {A→BE, E→C, C→AB} 2
Comment whether it satisfies 2NF or not with reason. If no, redesign the schema satisfying the properties of 2NF.

(b) Consider the relational schema Student_Performance(Name, CourseNo, RegNo, Grade) has the following functional dependencies: F = { (Name, CourseNo) → Grade, (RegNo ,CourseNo) → Grade, Name→ RegNo, RegNo→ Name} 2
Comment whether it satisfies BCNF or not with reason. If no, redesign the schema satisfying the properties of BCNF.
(GATE 2019)

(c) Consider the relation schema R(A, B, C, D) with functional dependency set F= { A↠D, A→B, A→C} Comment whether it satisfies 4NF or not with reason. If no, redesign the schema satisfying the properties of 4NF. 2

6. (a) Consider the following relational schema, employee (person-name, street, city), works (person-name, company-name, salary), Company (company-name, city), where the primary keys are underlined. Write a relational algebra query to find the names, street address, and cities of residence of all employees who work for TCS and earn more than 40,000 per annum. 2

(b) Write a query using Tuple relational calculus to find the name and company name of the employee who live in the same city as the company for which they work. 2

(c) Write a query using Domain relational calculus to display the name and salary of the employees working in the company placed in Delhi and employees belong to Mumbai. 2

7. (a) Assume that the companies may be located in several cities. Write a relational algebra query to find all company name located in every city where Infosys is located. (consider the schema given in 6(a)) 2

(b) Consider two relations R1(A,B,D) with tuples (1,5,2), (3,7,4), and R2(A,C) = (1,7), (4,9). Find out the left and right outer join of R1 and R2. 2

(c) Consider the following relation R 2

A	B	C	D
α	α	10	7
α	β	15	7
β	β	12	5
β	β	23	10

Write down the output of the following command.

i) $\pi_{A,B}((\sigma_{C>10}(R))$ ii) $\pi_{A,B}(\sigma_{A=B \wedge D>5}(R))$

8. (a) Explain how each of the ACID properties ensures reliable transaction processing in DBMS. 2

(b) Let Ri(z) and Wi(z) denote read and write operations on a data item z by a transaction Ti, respectively. Check whether the given schedule S with three transactions is a conflict serializable schedule or not. 2

S: R1(x) R2(x) W1(x) R3(x) W2(x).

(c) Let Ri(z) and Wi(z) denote read and write operations on a data item z by a transaction Ti, respectively. Check whether the given schedule S1 with four transactions is conflict serializable and recoverable or not. Justify your answer. 2

S1: R2(A) W3(A) Commit(T3) W1(A) Commit(T1) W2(B) R2(C) Commit (T2) R4(A) R4(B) Commit(T4). **(GATE 2014)**

9. (a) Prove or disprove: Every recoverable schedule is cascadeless. Justify your answer with an example. 2

(b) Consider schedule S with transaction T1 and T2. T1 transfer Rs.500 from account A to B, and T2 adds Rs.100 into account A. Prepare concurrent schedule with two phase locking protocol. 2

(c) The transactions T1, T2 with time stamps (TS) 5, 10 respectively want to write data item X. The current read time stamp is RTS(X) = 8 and write time stamp is WTS(X) = 7. Will the write be allowed for both the transactions T1, T2? Justify your answer. 2

10 (a) Given the following transactions and their timestamps, simulate the timestamp ordering protocol and determine whether each operation is allowed or leads to a rollback. 2
Initial: RTS(X)=0, WTS(X)=0

T1: TS = 10	**T2: TS = 20**
Read(X)	
	Read(X)
	Write(X)
Write(X)	

(b) Write down two major differences between immediate and deferred database modification. 2

(c) What actions will be taken by the log based recovery technique for deferred and immediate database modification, if a failure occurs after the Write(B) operation in the following schedule: 2

T1:	**T2:**	**T3:**
Read(A)		
A=A-500		
	Read(B)	
	B=B+100	
Write(A)		
Commit		
		Read(C)
	Write(B)	
	Commit	

End of Questions